

5 Sulfur

Sulfur is part of amino acids, proteins, enzymes, and plant defenses. Sulfate is mobile in soil and may enter from fertilizer, gypsum, organic matter, and irrigation water.

Why it matters

- **Protein and enzymes.** Sulfur is needed to make sulfur-containing amino acids and many enzymes.
- **Multiple sources.** Crop supply can come from organic matter, sulfate fertilizers, gypsum, atmospheric inputs, and irrigation water.

Tentative leaf guide

Measure	Interpretation note
UC table requires caution	The published avocado guide shows an overlap: less than 0.5 percent in the deficient column and 0.20 to 0.60 percent as adequate, with more than 1.0 percent as excess.
Practical use	Because the printed categories overlap, interpret S with the laboratory and advisor, multi-year trends, source history, and other root-zone evidence.

Main soil and environmental controls

Factor	What it can mean in the root zone
Microbes and organic matter	Microbes release sulfate as organic matter decomposes and may temporarily hold S in their cells.
pH	Elemental S can acidify soil only after microbial oxidation. The process needs time, oxygen, moisture, and suitable temperature and can add salinity.
Texture and leaching	Sulfate moves with water and can leach readily from sandy or low-holding-capacity soil.
Irrigation water	Sulfate in water contributes to both S supply and total salinity. Include it in the salt and nutrient budget.
Roots and drainage	Dry soil limits movement; waterlogging limits roots and microbial oxidation. Leaching requires a drainage outlet.

Field clues and checks

- **Inventory all sources.** Record sulfate in water, gypsum, blended fertilizers, compost, manure, and acidifying products.
- **Do not use elemental S as a quick correction.** The pH response is gradual and strongly depends on soil buffering and biological conditions.
- **Watch EC.** Sulfate fertilizers and gypsum add dissolved ions even when they serve another purpose.

DST interpretation The Phase 1 K by S interaction was robust enough for research follow-up but varied by year and nutrient background. It should guide measurement of sulfate, water salinity, and fertilizer history—not a K or S prescription.